CSD404 Internet of Things – Case Study 1

Instructions:

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Due Date: 11 am, Feb 5, 2021

Submitting this Assignment

You will submit (upload) this assignment in Blackboard. Email submissions will not be accepted.

- Write your answer after the given question in this document.
- Name this document as CA1_IOT2021_John_Doe.pdf in case your name is John Doe.

Grading Criteria

This assignment has 3 points (with **weightage of 3%** in your overall 100 points)

Question

Paper Name (with all details):

Geoffrey Werner-Allen, Konrad Lorincz, Matt Welsh, Omar Marcillo, Jeff Johnson, Mario Ruiz, and Jonathan Lees. 2006. Deploying a Wireless Sensor Network on an Active Volcano. *IEEE Internet Computing*, 10, 2 (March 2006), 18–25. DOI:https://doi.org/10.1109/MIC.2006.26

Summarize the paper (100-150 words):

WSNs with numerous resource-limited nodes which are linked with one another via low-bandwidth wireless radios have been at the center of WSN research for quite some time. They have shown great potential in terms of data collection and supporting scientific studies and have taken data collection to a new scale which was not seen previously. One of the areas that has benefitted from this progress is the geophysical study of volcanic activity, something which was considered an arduous task before.

In the bid to collect data from volcanoes, the sensors need to be designed in a particular format. This has been discussed in detail in the paper. These include high data rates and high fidelity rates and triggered event detection capabilities. Their work leads to many exciting future avenues for research of geophysical studies by deployment in areas humans have not been able to access before.

In the lecture, we discussed various characteristic features and design requirements of a sensor network. Which of those features and requirements are addressed for the given application; also write how they have been addressed. (500-600 words):

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The authors have stated many requirements for a wireless sensor network to be able to work in volcanic studies which resonate with the discussions we have had in class. These have been addressed in the following section:-

Reliability is an important aspect of these wireless sensor networks and it becomes a challenge when collecting high-fidelity data due to two major reasons. One, the radio links in these locations are lossy and frequently asymmetrical and two, the low-cost crystals used in the oscillators of these nodes have a low fault tolerance which leads to varying clock rates. It is pertinent to have a fault tolerant system at a volcano because of the extreme conditions that the sensors have to go through and the quality of the data might be affected if they are not reliable at these extreme conditions.

The redressal mechanism for the same was novelly developed by the authors and the data collection prototype was called Fetch which allowed them to collect data from each node over a multihop network. They relied on chunk processing which was locally buffered in blocks of 256 bytes and then tagged with sequence numbers and time stamps. These chunks are collected at regular intervals to make sure that the last relayed message is with the data retrieval team.

The **topology** was also an important aspect of these networks because how they would be set up would affect data quality and retrieval. The stations were installed in a linear fashion that radiated away from the volcanoes vent and even produced an aperture of three kilo metres. The stations were positioned as far away from each other as possible depending on how the radios of each node would permit. The terrain of the region often forced them to install certain stations close to each other even though they could maintain links of almost 400 metres. This was done in a trial and error format because at times it was noticed the the geography was such that the station was communicating with a node which wasn't the one they were expecting (closest one) but instead communicated with the node far away which would lead to them removing certain station . In addition to the nodes, they also used FreeWave radio modems, charged using a car battery, to provide a long-distance and reliable radio link between the sensor network and observatory laptop.

In contrast to the existing volcanic data collection tools, the nodes developed by the authors are lighter in weight, smaller in size and **consume less power.** This was an important feature which the authors focused on to create low-cost, power efficient and volcanic study system. The spatial distribution that they created also leads to a low power consumption over a period of time and hence, have successfully collected data from Mt. Reventador. Their low cost and power consumption nodes also make it **scalable** to other projects. The authors themselves spread the network over 3km recording over 230 events with ease. The scalability in terms of this application is only how many events can the sensor capture reliably because this is a local study at the end of the day. The materials that the authors use to build and power the nodes are what makes it **low cost.** They use a pair of alkaline D cell batteries which power each sensor node remote at the location of the node itself which maximize the node lifetime and reduce the cost.

The **network hardware** comprised of 16 stations equipped with seismic and acoustic

sensors. There were some constraints with this as well which include selecting the microcontroller. The authors couldn't use the analog-to-digital converters on the MSP430 microcontroller because they only provide 16 bits of resolution while the authors needed atleast 20 bits. Secondly, seismo-acoustic signals often require an aggressive filter centered around 50 Hz but implementing such a filter using analog components wasn't feasible. Hence, they used additional TI ADCs and made a custom hardware board.

Another aspect that the authors focused on was **overcoming high data rates during event detection** but low during transmission. This leads to a buffering because the network is collecting data much faster than it is relaying it to the storage apparatus. The authors used the TMote Sky's flash memory which can fill in 20 minutes of recording at almost 100 Hz. Luckily for authors this window was enough to collect reliable data and hence, all they had to focus on was timing the data collection window properly.